1. **Given the sales data of each quarter in three years, please calculate the seasonal index in the table and the corresponding deseasonalized data?**

Solution:

Quarter	Actual	Seasonal Index	Deseasonalized Data
2015Q1	154	1.075044	141.6
2015Q2	126	0.879581	141.4
2015Q3	108	0.753927	145.5
2015Q4	185	1.291449	144.6
2016Q1	158	1.110721	145.3
2016Q2	137	0.963093	153.7
2016Q3	105	0.738137	141.5
2016Q4	169	1.188049	132.1
2017Q1	145	1.076067	133.4
2017Q2	112	0.831169	125.7
2017Q3	99	0.734694	133.4
2017Q4	183	1.358071	143.1

What is the sum of the seasonal indices and the average of the seasonal indices, respectively?

Solution:

The sum of the seasonal indices is "the number of seasons", while the average of the seasonal indices is always 1.

What would the seasonal indices for Q2 2016 and Q3 2016 be if they are the same?

Solution: 1.3

2. **Given the sequence data below, calculate the support of 1-sequences.**

ID	Sequences

Seq 1	**{a,b}{c}{f}{g}{e}**
Seq 2	**{a,d}{c}{b}{a,b,e,f}**
Seq 3	**{a}{b}{f}{e}**
Seq 4	**{b}{f,g}**

Solution:

1-Sequence	Support Count
{a}	3
{b}	4
{c}	2
{d}	1
{e}	3
{f}	4
{g}	2

Given the min_support = 50% (mini_support count=2), find out all frequent 1-sequential patterns. Based on the frequent 1-sequential patterns, how many candidate 2 sequences can be generated? Please write down all candidate 2 sequences.

Solution: Given the min_support_count=2, then {d} is infrequent. The remaining frequent 1 sequences can generate 51 candidate 2-sequences and they are as follows:

	a	b	c	e	f	g
a	<{a}{a}>	<{a}{b}><{ab}>	<{a}{c}><{ac}>	<{a}{e}><{ae}>	<{a}{f}><{af}>	<{a}{g}><{ag}>
b	<{b}{a}>	<{b}{b}>	<{b}{c}><{bc}>	<{b}{e}><{be}>	<{b}{f}><{bf}>	<{b}{g}><{bg}>
c	<{c}{a}>	<{c}{b}>	<{c}{c}>	<{c}{e}><{ce}>	<{c}{f}><{cf}>	<{c}{g}><{cg}>
e	<{e}{a}>	<{e}{b}>	<{e}{c}>	<{e}{e}>	<{e}{f}><{ef}>	<{e}{g}><{eg}>
f	<{f}{a}>	<{f}{b}>	<{f}{c}>	<{f}{e}>	<{f}{f}>	<{f}{g}><{fg}>

g	<{g}{a}>	<{g}{b}>	<{g}{c}>	<{g}{e}>	<{g}{f}>	<{g}{g}>

3. List all the 4-subsequences contained in the following data sequence:<{1,3}{2}{2,3}{4}>.

Solution (14 subsequences):

< {1, 3} {2} {2} >	< {1, 3} {2} {3} >
< {1, 3} {2} {4} >	< {1, 3}{2, 3} >
< {1, 3} {3} {4} >	< {1} {2} {2, 3} >
< {1} {2} {2} {4} >	< {1} {2} {3} {4} >
< {1} {2, 3} {4} >	< {3} {2} {2, 3} >
< {3} {2} {2} {4} >	< {3} {2} {3} {4} >
< {3} {2, 3} {4} >	< {2} {2, 3} {4}>

List all the 3-element subsequences contained in the data sequence: <{1,3}{2}{2,3}{4}>.

Solution (21 subsequences):

< {1, 3} {2} {2, 3} >	< {1, 3} {2} {4} >
< {1, 3} {3} {4} >	< {1, 3} {2} {2} >
< {1, 3} {2} {3} >	< {1, 3} {2, 3} {4} >
< {1} {2} {2, 3} >	< {1} {2} {4} >
< {1} {3} {4} >	< {1} {2} {2} >
< {1} {2} {3} >	< {1} {2, 3} {4} >
< {3} {2} {2, 3} >	< {3} {2} {4} >
< {3} {3} {4} >	< {3} {2} {2} >
< {3} {2} {3} >	< {3} {2, 3} {4} >
<{2} {2,3} {4}>	<{2} {2} {4}>
<{2} {3} {4}>	